PIYUSH SATTI

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SUMMARY

Full-stack software developer with 2 years of hands-on experience building and shipping production applications, owning backend services, data models, and user-facing workflows.

TECHNICAL SKILLS

Languages	Python, JavaScript/TypeScript, Java, SQL, HTML/CSS
Full-stack	React, Redux FastAPI, Flask, Spring Boot, Express/Node, REST, GraphQL
Agentic Systems	LLM APIs, Prompt Engineering, RAG pipelines, LangGraph, Vector Databases
Tools & Storage	PostgreSQL (Supabase), MongoDB, Redis, Git/GitHub, Docker, GitHub Actions, CI/CD

EDUCATION

Concordia University (GPA: 3.62/4.3) — Montreal, Quebec, Canada
Master of Science, Applied Computer Science — *Sept. 2023 – Aug. 2025*

Thapar Institute of Engineering and Technology (GPA: 8.96/10) — Patiala, Punjab, India
Bachelor of Engineering, Electronics and Computer Engineering — *Jun. 2017 – Jun. 2021*

EXPERIENCE

Teaching Assistant — Montreal, Quebec, Canada
Concordia University — *Jan. 2025 – Apr. 2025*

- Programmer on Duty (Java) for Object-Oriented Programming. Taught OOP concepts, File I/O, Polymorphism, Recursion, Exception Handling, Abstract Classes and Interfaces, and Inheritance.

Software Developer — Remote
Freelance (Upwork, forums) — *Aug. 2021 – Apr. 2023*

- Delivered end-to-end client software engagements from requirements discovery through design, implementation, feedback iterations, and handoff, consistently working under fixed scope and deadlines.
- Built backend services and third-party API integrations in Python (FastAPI, Flask), focusing on maintainable code, correctness, and operational stability.
- Developed a vision-based game QA automation harness using OpenCV for on-screen state detection and PyAutoGUI for input control, enabling repeatable regression runs.
- Shipped an ML-powered movie recommendation web app with a React frontend and a Node.js backend, exposing APIs for search and personalized recommendations.

PROJECTS

Community Event Signup & Approval Platform — github.com/piyushsatti/nonagon
Python, JavaScript, React, FastAPI, GraphQL, PostgreSQL, Supabase — *Nov. 2025 – Present*

- Built a **GraphQL-first full-stack platform** for community event postings and sign-up workflows, implementing schema-driven queries and mutations on FastAPI (Strawberry) with a React client.
- Implemented **username/password authentication** with secure password hashing and **JWT-based sessions**, plus resolver-level RBAC to secure approvals, edits, and visibility rules.
- Modeled core workflows in **PostgreSQL** with SQLAlchemy, enforcing referential integrity and pagination patterns enabling infinite-scroll views across events, sign-up requests, approvals, and notifications.
- Delivered a **dashboard and analytics UI** and an interactive **relationship graph view** (zoom, pan, scroll) in React to visualize linked entities and surface engagement signals.

Turn-Based Strategy Game Engine & Map Editor — github.com/piyushsatti/risk-emulated
Java, JUnit — *Jan. 2024 – Apr. 2024*

- Built a **turn-based strategy game engine** with map-editor and gameplay modes, orchestrating player setup, country assignment, reinforcement calculation, and round-robin order execution.
- Implemented a **State**-driven phase controller and **Command**-style parsing/validation to gate actions by phase and convert terminal input into structured operations.
- Developed a **graph-based map builder and validator** enforcing world connectivity and continent constraints to ensure only legal maps are loadable and playable.
- Structured the codebase with **MVC separation** and **Strategy**-based policy hooks to support pluggable AI behaviors and targeted JUnit test scenarios.